

Reviewer Pack — Minimal p-Adic Logarithmic Potential and Communication Compl...

Entry 146afffa99fc · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal p-Adic Logarithmic Potential and Communication Complexity Rank

Entry ID: 146afffa99fc **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 08:39:56 UTC

1. Conjecture

Field A (mathematical branch): p-Adic Analysis **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For all satisfiability instances with $n \leq 40$ variables, the communication complexity rank (CR) is linearly correlated with the p-adic logarithmic potential (PLP) of the associated boolean function, such that $CR(\varphi_n) = \Theta(PLP(\varphi_n))$ for every formula φ_n .

Rationale (proposer's reasoning):

p-Adic analysis provides a framework to study arithmetic properties of functions. The correlation between communication complexity and p-adic logarithmic potential could reveal underlying arithmetic structures in boolean functions that are not apparent through standard complexity measures, potentially leading to new insights into communication complexity lower bounds.

Taxonomy category: p-adic_analysis_communication_complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): bb8e8458b3b4e436

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between Communication Complexity Rank (CR) and p-adic Logarithmic Potential (PLP) is ≥ 0.7 for at least 25 out of 30 seeds, with no seed showing a correlation ≤ 0.5 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.70	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): -p-adic analysis AND communication complexity rank-boolean function p-adic logarithmic potential AND linear correlation-satisfiability instances communication complexity rank p-adic analysis

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
from fractions import Fraction

def generate_cnf(n):
    cnf = []
    for _ in range(10 * n): # Ensure enough clauses to cover all variables
        num_clauses = random.randint(1, n)
        clause = [random.choice([-i, i]) for i in range(1, n + 1) if random.random() < 0.5]
        cnf.append(clause)
    return cnf

def compute_plp(cnf):
    # Placeholder implementation of p-adic logarithmic potential
    # This is a dummy function and should be replaced with actual computation
    return sum(len(clause) for clause in cnf)

def compute_cr(cnf):
    # Placeholder implementation of communication complexity rank
    # This is a dummy function and should be replaced with actual computation
    return len(cnf)

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 40
    cnf = generate_cnf(n)
    plp = compute_plp(cnf)
    cr = compute_cr(cnf)

    correlation_coefficient = (cr * plp) / (abs(cr) * abs(plp)) if cr != 0 and plp != 0 else 0

    return {
        "metric_name": "correlation_coefficient",
```


relation coefficient calculation is based on these placeholders, making it impossible to confirm or challenge the conjecture with this data.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The critic challenged the validity of the test due to placeholder functions for p-adic logarithmic potential (PLP) and communication complexity rank (CR), which are not implemented correctly. This makes it impossible to confirm or challenge the conjecture with this data. | next: Implement correct PLP and CR functions and retest the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11956
2	preregistration	ollama_remote	glm4:latest	0	0	9232
3	novelty	ollama_remote	glm4:latest	0	0	8326
4	novelty	ollama_remote	glm4:latest	0	0	9162
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17032
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12001
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14114
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10350
9	critic	ollama_remote	glm4:latest	0	0	22470
10	judge	ollama_remote	glm4:latest	0	0	9386

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 124029 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/146afffa99fc.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/146afffa99fc.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/146afffa99fc.tar.gz (if generated)